

Nonexistence of Simple Graphs with Laplacian Spectrum $\{0, 1, \dots, n - 1\}$ for $n = 16$ and 20 : Exact Computer-Assisted Proofs

Preprint, version 0.1.0

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24 August 2026

MSC 2020: 05C50, 05C85, 15A18.

Abstract

We establish the $n = 16$ and $n = 20$ cases of the $S_{n,n}$ conjecture. There is no simple graph on 16 vertices with Laplacian spectrum $\{0, 1, \dots, 15\}$, and no simple graph on 20 vertices with Laplacian spectrum $\{0, 1, \dots, 19\}$. Both proofs combine exact enumeration of forced degree data with linear relaxations derived from spectral moments, graph and complement identities, and degree-class incidence constraints. The order-16 proof eliminates 2,077 candidate degree sequences with 2,143 exact rational Farkas leaves organized into degree-reduction certificates and 42 integer branch trees. The order-20 proof enumerates 160,244 degree-sequence/triangle-count pairs, reduces them to 80,122 complement representatives, and eliminates them with 80,124 exact integer Farkas certificates, including four exhaustive branch leaves. All certificate checks use integer or rational arithmetic, and both coverage ledgers have zero uncovered cases. These finite-order results do not settle the conjecture in general.

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Introduction and relation to previous work

For $1 \leq i \leq n$, Fallat, Kirkland, Moliterno and Neumann introduced

$$S_{i,n} = \{0, 1, \dots, n\} \setminus \{i\}$$

and asked which such sets occur as the Laplacian spectrum of a simple graph [1]. Their $S_{n,n}$ conjecture states that

$$S_{n,n} = \{0, 1, \dots, n-1\}$$

is not Laplacian-realizable for any $n \geq 2$. Fallat et al. established the conjecture for $n \leq 11$, prime n , and $n \equiv 2, 3 \pmod{4}$ [1]. Das, Lee and Cheon derived additional diameter restrictions on any realizing graph and its complement [2]. Goldberger and Neumann proved the conjecture for all sufficiently large orders [3]. Hameed, Khan and Tyaglov surveyed the remaining range and, among other structural results, excluded Cartesian-product realizations [4]. Johnston, Plosker and Varona subsequently established the unrestricted order-12 case and described $n = 16$ as the smallest open order in July 2026 [5].

The present paper treats exactly two orders, 16 and 20. Within the finite cases and order classes covered by the cited results, they are the two smallest orders not covered: 17 and 19 are prime, while $18 \equiv 2 \pmod{4}$. The least order not covered by that cited collection after the present two theorems is 21. The proofs share a mathematical architecture but not a single certificate corpus. The order-16 computation uses rational Farkas certificates and complete integer branch trees for 42 final class models. The order-20 computation uses a complement quotient followed by three continuous relaxations and a final two-root integer split.

A literature search completed on 24 August 2026 found no earlier public proof of either case.

1. Main results and proof architecture

Let G be a finite simple undirected graph, let A be its adjacency matrix, let $D = \text{diag}(d_v)$, and let $L = D - A$ be its Laplacian.

Theorem 1 (order 16)

There is no simple undirected graph G on 16 vertices such that

$$\text{spec } L(G) = \{0, 1, 2, \dots, 15\}.$$

Theorem 2 (order 20)

There is no simple undirected graph G on 20 vertices such that

$$\text{spec } L(G) = \{0, 1, 2, \dots, 19\}.$$

Each proof has four logical components:

1. derive necessary spectral, degree and triangle conditions;
2. enumerate every discrete candidate satisfying those conditions;
3. map every hypothetical target graph to a feasible point of a sequence of necessary linear or mixed-integer relaxations;
4. verify exact Farkas certificates, together with exhaustive integer branch splits where required, that eliminate every candidate.

Floating-point optimization was used only during discovery. The proofs rely on the analytic derivations, exact enumeration, certificate verification and coverage ledgers described below.

Appendix A gives the complete enumeration and coverage ledgers. Appendices B and C give the variables, rows, bounds, serialization and certificate semantics used by the order-16 and order-20 verifiers.

2. Shared spectral and graph identities

Assume temporarily that G has n vertices and Laplacian spectrum $\{0, 1, \dots, n-1\}$. The zero eigenvalue is simple, so G is connected. The first three spectral power sums give

$$\sum_v d_v = \text{tr } L = \frac{n(n-1)}{2}, \quad (2.1)$$

$$\sum_v d_v^2 = \text{tr } L^2 - \text{tr } L = \frac{n(n-1)(n-2)}{3}, \quad (2.2)$$

and

$$\sum_v d_v^3 - 6t(G) = \text{tr } L^3 - 3 \sum_v d_v^2. \quad (2.3)$$

Here we used

$$(L^2)_{vv} = d_v^2 + d_v, \quad \text{tr } L^3 = \sum_v d_v^3 + 3 \sum_v d_v^2 - 6t(G).$$

The complement satisfies

$$L(\overline{G}) = nI - J - L(G).$$

On $\mathbf{1}^\perp$, eigenvalue k is sent to $n-k$. Hence the target spectrum is complement-invariant. Direct counting also gives

$$t(G) + t(\overline{G}) = \binom{n}{3} - \frac{1}{2} \sum_v d_v(n-1-d_v) = \frac{n(n-1)(n-5)}{12}. \quad (2.4)$$

For a local degree bound, choose an orthonormal Laplacian eigenbasis u_k and put $w_{v,k} = u_k(v)^2$. Connectedness gives $w_{v,0} = 1/n$, while

$$\sum_k k w_{v,k} = d_v, \quad \sum_k k^2 w_{v,k} = d_v^2 + d_v. \quad (2.5)$$

For $1 \leq k \leq n-1$,

$$k^2 \leq nk - (n-1)$$

because $(k-1)(n-1-k) \geq 0$. Substitution in (2.5) gives

$$nd_v^2 - n(n-1)d_v + (n-1)^2 \leq 0. \quad (2.6)$$

This proves the exact integer ranges used below:

$$2 \leq d_v \leq 13 \quad (n = 16), \quad 2 \leq d_v \leq 17 \quad (n = 20). \quad (2.7)$$

Finally, every triangle contributes at each of its three vertices, so

$$3t(G) \leq \sum_v \binom{d_v}{2}.$$

Combining this bound for G and $\bar{G}$ with (2.4) yields

$$54 \leq t(G) \leq 166 \quad (n = 16), \quad 127 \leq t(G) \leq 348 \quad (n = 20). \quad (2.8)$$

These bounds are consequences of the target spectrum, not imported pruning assumptions.

3. Common relaxation variables

Let m_d be the multiplicity of degree d . For an order- n candidate define the nonnegative real spectral weights

$$Y_{d,k} = n \sum_{v:d_v=d} u_k(v)^2, \quad 1 \leq k \leq n-1. \quad (3.1)$$

Every target graph satisfies

$$\sum_k Y_{d,k} = (n-1)m_d, \quad \sum_k kY_{d,k} = nm_d d, \quad \sum_k k^2 Y_{d,k} = nm_d(d^2 + d), \quad (3.2)$$

and, for each k ,

$$\sum_d Y_{d,k} = n. \quad (3.3)$$

For a vertex v of degree d , put

$$s_v = \sum_{u \sim v} d_u, \quad q_v = \#\{\text{triangles containing } v\}, \quad h_v = s_v - 2q_v.$$

Direct multiplication gives

$$(L^3)_{vv} = d^3 + 2d^2 + h_v. \quad (3.4)$$

If $R_v = V \setminus (N(v) \cup \{v\})$, then

$$h_v = d + e(N(v), R_v), \quad d \leq h_v \leq d(n-d). \quad (3.5)$$

For a degree class define

$$S_d = \sum_{v:d_v=d} s_v, \quad H_d = \sum_{v:d_v=d} h_v.$$

Equations (3.1) and (3.4) imply

$$\sum_k k^3 Y_{d,k} = n\{m_d(d^3 + 2d^2) + H_d\}. \quad (3.6)$$

Summing (3.5) over the degree- d class gives the two necessary Stage-2 inequalities

$$m_d d \leq H_d \leq m_d d(n - d). \quad (3.7)$$

In both serialized Stage-2 models, H_d is not a separate variable. It is the affine abbreviation

$$H_d = \frac{1}{n} \sum_k k^3 Y_{d,k} - m_d(d^3 + 2d^2)$$

obtained from (3.6). The variables S_d are allowed to be real in the relaxation, even though the values supplied by an actual graph are integers. After eliminating H_d , (3.7) and the bounds below are linear rows in Y, S .

The neighbor-degree order bounds, triangle capacities, complement-triangle identity and $\sum_d S_d = \sum_v d_v^2$ complete the Stage-2 relaxation. If ℓ_d, u_d are the sums of the d smallest and largest entries after removing one occurrence of d , then

$$m_d \ell_d \leq S_d \leq m_d u_d, \quad (3.8)$$

while

$$H_d \leq S_d \leq H_d + 2m_d \binom{d}{2}. \quad (3.9)$$

Writing $M = n(n-1)/4$ for the forced edge count, the local complement formula

$$\bar{q}_v = \binom{n-1-d}{2} - M + s_v - q_v \quad (3.10)$$

gives

$$m_d \left(2M - 2 \binom{n-1-d}{2} \right) \leq S_d + H_d \leq 2M m_d. \quad (3.11)$$

All these rows are necessary conditions. The models deliberately omit several nonlinear graph-realizability constraints; infeasibility of a relaxation is therefore sufficient for nonexistence, whereas feasibility is not evidence of a graph.

4. The order-16 certified exhaustion

4.1 Forced data and exact enumeration

For $n = 16$, (2.1)–(2.3) specialize to

$$\sum_v d_v = 120, \quad \sum_v d_v^2 = 1120, \quad \sum_v d_v^3 - 6t = 11040. \quad (4.1)$$

The complement triangle total is 220, the degree range is $[2, 13]$, and the triangle range is $[54, 166]$. Exact nondecreasing-sequence enumeration has the following ledger.

Order-16 layer	Source	Eliminated	Survivors
moment equations	–	–	2,249
Erdős–Gallai graphicality	2,249	16	2,233
integral triangle count and bounds	2,233	156	2,077
Stage 1 spectral weights	2,077	1,359	718
Stage 2 local third moments	718	676	42
integer class branch trees	42	42	0

A second implementation enumerates the moment solutions by degree multiplicities and checks graphicality by Havel-Hakimi instead of reusing the main combinations/Erdős–Gallai route. It reproduces 2,249, 2,233 and 2,077 exactly.

The recurrence, pruning invariant, graphicality test, triangle filter, deterministic record order and exact ledger-partition obligations for this enumeration are specified in Appendix A.

4.2 Spectral and local-third-moment leaves

The Stage-1 system consists of (3.2)–(3.3) at $n = 16$. Exact rational Farkas certificates eliminate 1,359 candidates. Stage 2 appends the real variables S_d , uses the aggregate bounds (3.7)–(3.9) and (3.11) with H_d eliminated by the affine expression obtained from (3.6), and adds

$$\sum_d S_d = 1120.$$

Exact rational Farkas certificates eliminate another 676 candidates, leaving 42 sequences.

4.3 Integer degree-class model

For a surviving degree multiset let D be its distinct degrees and put

$$P_{a,b} = \begin{cases} \binom{m_a}{2}, & a = b, \\ m_a m_b, & a < b. \end{cases}$$

The model retains the real variables $Y_{d,k}$ and adds integer graph counts:

- $E_{a,b}$, the number of graph edges with endpoint degree multiset $\{a, b\}$;
- $T_{a,b,c}$, the number of graph triangles with degree multiset $\{a, b, c\}$;
- $C_{a,b,c}$, the analogous complement-triangle count indexed by original degrees.

Their bounds are the elementary pair and triple capacities. Stub equations, graph and complement triangle totals, classwise third-moment equations, edge-triangle incidence bounds and degree-class wedge bounds are imposed. For example,

$$2E_{d,d} + \sum_{b \neq d} E_{\min(d,b), \max(d,b)} = m_d d, \quad (4.2)$$

$$\sum T_{a,b,c} = t, \quad \sum C_{a,b,c} = 220 - t, \quad (4.3)$$

and, with $\nu_d(a, b, c)$ the multiplicity of d in a triple,

$$\sum_k k^3 Y_{d,k} = 16 [m_d(d^3 + 2d^2) + S_d^G - 2R_d^G], \quad (4.4)$$

where

$$S_d^G = 2dE_{d,d} + \sum_{b \neq d} bE_{\min(d,b), \max(d,b)}, \quad R_d^G = \sum_{a \leq b \leq c} \nu_d(a, b, c) T_{a,b,c}.$$

The complement uses $P_{a,b} - E_{a,b}$, original degree d maps to $15 - d$, and spectral weights satisfy

$$\bar{Y}_{15-d,k} = Y_{d,16-k}. \quad (4.5)$$

Any graph with the prescribed spectrum determines a feasible mixed point: Y is generally real, whereas E, T, C are integer. The converse is not asserted. The complete triple set, capacities, graph and complement rows, wedge and incidence inequalities, variable order and branch-tree serialization are given in Appendix B; that appendix is the formal definition of the model whose certificates are checked here.

4.4 Exact branch-and-Farkas proof

At an order-16 branch-tree node, the model has rational rows

$$Ax \leq b, \quad Cx = e.$$

A leaf stores rational multipliers y, z satisfying

$$y \geq 0, \quad y^\top A + z^\top C = 0, \quad y^\top b + z^\top e = -1. \quad (4.6)$$

Any feasible x would yield $0 \leq -1$, a contradiction. An internal node branches on an explicitly integer variable x_j using

$$x_j \leq a \quad \text{or} \quad x_j \geq a + 1. \quad (4.7)$$

The cases are disjoint and exhaustive for every integer value in the parent interval. The 42 trees contain 174 nodes, 66 branch nodes and 108 exact Farkas leaves, with maximum depth 10. Together with the 1,359 + 676 earlier leaves, the order-16 proof verifies 2,143 exact Farkas certificates. No target candidate survives the certified computation. Lemma 3 below supplies the graph-to-model implication that completes the proof of Theorem 1.

Computational Proposition 1 (order-16 exact coverage)

The order-16 ledger partitions all 2,077 candidates into 1,359 Stage-1 leaves, 676 Stage-2 leaves and 42 class roots. The branch forest replaces those roots by 108 certified leaves, so 2,143 rational Farkas certificates cover every candidate.

5. The order-20 certified exhaustion

5.1 Forced data, enumeration and complement quotient

For $n = 20$, the shared identities specialize to

$$\sum_v d_v = 190, \quad \sum_v d_v^2 = 2280, \quad \sum_v d_v^3 - 6t = 29260. \quad (5.1)$$

The complement triangle total is 475, the degree range is $[2, 17]$, and the triangle range is $[127, 348]$. Exact enumeration gives

$$209,932 \longrightarrow 200,108 \longrightarrow 160,244 \quad (5.2)$$

moment sequences, graphical sequences, and graphical degree-sequence/triangle-count pairs. The primary enumerator lists degree multiplicities and uses Erdős–Gallai. A position-by-position/Havel–Hakimi audit and a degree-alphabet meet-in-the-middle/Kleitman–Wang audit reproduce the same counts and complete uncompressed candidate bytes.

The exact enumeration recurrences, pruning invariants, record serialization, complement-orbit checks and ledger partitions are specified in Appendix A.

For a pair (s, t) , where $s = (d_1, \dots, d_{20})$ is nondecreasing, define

$$\kappa(s, t) = (\text{sort}(19 - d_1, \dots, 19 - d_{20}), 475 - t). \quad (5.3)$$

The map is a fixed-point-free involution on the 160,244 candidates: a fixed point would require $2t = 475$. Retaining the lexicographically smaller member of each orbit leaves exactly 80,122 representatives. If a target graph realizes either member, it or its complement realizes the retained one.

5.2 Three necessary relaxation layers

Stage 1 is (3.2)–(3.3) at $n = 20$. Stage 2 appends the real variables S_d , uses the aggregate bounds (3.7)–(3.9) and (3.11) with H_d eliminated by the affine expression obtained from (3.6), and adds $\sum_d S_d = 2280$. Stage 3 uses the real spectral weights together with degree-pair and degree-triple variables $E_{a,b}, T_z, C_z$. For a nondecreasing triple z with degree multiplicities $\nu_d(z)$, put

$$P_z = \prod_d \binom{m_d}{\nu_d(z)}.$$

Only triples with $P_z > 0$ are serialized. Define the pair multiplicity in a triple by

$$\pi_{a,b}(z) = \begin{cases} \binom{\nu_a(z)}{2}, & a = b, \\ \nu_a(z)\nu_b(z), & a < b. \end{cases} \quad (5.4a)$$

The elementary capacities are

$$0 \leq E_{a,b} \leq P_{a,b}, \quad 0 \leq T_z, C_z \leq P_z. \quad (5.4)$$

The model imposes stub equations, graph and complement triangle totals, graph and complement classwise L^3 equations, pair-incidence bounds and wedge bounds. For a pair (a, b) , graph triangles incident with that pair satisfy

$$\sum_z \pi_{a,b}(z) T_z \leq (\min(a, b) - 1) E_{a,b}, \quad (5.5)$$

while complement triangles satisfy

$$\sum_z \pi_{a,b}(z) C_z \leq (18 - \max(a, b))(P_{a,b} - E_{a,b}). \quad (5.6)$$

In the 343 ordinary Stage-3 leaves, E, T, C are allowed to be real. Thus each certificate proves infeasibility of a continuous relaxation containing every actual graph point. Integrality metadata is consulted only for the final branch variable described below.

Appendix C states the full graph and complement classwise equations, stub and triangle-total rows, pair-incidence and wedge inequalities, standard-form conversion, deterministic row order and sparse-certificate serialization reconstructed by the verifier.

5.3 Exact standard-form certificates

For each order-20 leaf, finite lower bounds are shifted away, inequalities and finite upper bounds receive nonnegative slack variables, and the verifier obtains

$$Bw = h, \quad w \geq 0, \quad (5.7)$$

with integer B, h . A stored integer multiplier z is a contradiction if

$$B^T z \geq 0, \quad h^T z < 0. \quad (5.8)$$

Indeed, a feasible w would imply

$$h^T z = w^T B^T z \geq 0.$$

The dedicated slack columns force the multipliers of inequality and upper-bound rows to be nonnegative. The verifier checks integer arithmetic, row signs, row order and every sparse multiplier exactly.

5.4 Coverage and final integer split

The exact order-20 coverage ledger is:

Order-20 layer	Source	Exact Farkas leaves	Survivors
complement representatives	80,122	—	80,122
Stage 1 spectral weights	80,122	63,235	16,887
Stage 2 local third moments	16,887	16,542	345
Stage 3 class model	345	343	2 roots
integer split	2 roots	4	0

The two parent roots are

$$s^{(0)} = (2, 2, 2, 4, 6, 6, 7, 7, 9, 9, 9, 10, 12, 12, 14, 14, 15, 16, 17, 17), \quad t^{(0)} = 229, \quad (5.9)$$

and

$$s^{(1)} = (2, 2, 2, 5, 5, 6, 6, 8, 9, 9, 9, 10, 12, 12, 14, 14, 15, 16, 17, 17), \quad t^{(1)} = 231. \quad (5.10)$$

Both have exactly three degree-2 vertices. Variable $E_{2,2}$ is an actual integer graph-edge count with parent interval $[0, 3]$. The two children

$$E_{2,2} \leq 0 \quad \text{and} \quad E_{2,2} \geq 1 \quad (5.11)$$

are disjoint and cover every possible integer value. All four child relaxations have exact Farkas certificates. Hence the number of exact order-20 leaves is

$$63,235 + 16,542 + 343 + 4 = 80,124. \quad (5.12)$$

Every source at each layer is the disjoint union of the certified leaves and the next survivor ledger. The verifier checks exact set difference, uniqueness, source order, shard hashes and zero uncovered cases. Therefore no retained candidate survives the certified computation. Lemma 4 below supplies the canonical complement and graph-to-model implications that complete the proof of Theorem 2.

Computational Proposition 2 (order-20 exact coverage)

The order-20 ledgers partition all 80,122 complement representatives into 63,235 Stage-1 leaves, 16,542 Stage-2 leaves, 343 Stage-3 leaves and two final roots. The four certified children of those roots complete the 80,124-leaf exhaustion.

6. Order-specific graph-to-model soundness and theorem proofs

The shared identities do not make the two finite computations identical. The order-16 proof branches through complete mixed-integer class trees, whereas the order-20 proof first selects one complement representative, uses continuous class relaxations, and consults integrality only for its final split. We therefore state the two mapping lemmas separately.

Lemma 3 (order-16 graph-to-model mapping)

Suppose that a simple graph G on 16 vertices has Laplacian spectrum $\{0, 1, \dots, 15\}$. Its sorted degree sequence and forced integral triangle count occur in the complete order-16 enumeration. Its projector-diagonal weights Y satisfy the Stage-1 equations, and the same weights together with the actual neighbor-degree sums S_d satisfy every Stage-2 row. If the candidate reaches one of the 42 class models, the same Y together with the actual integer degree-class counts E, T, C gives a feasible mixed point of that model. At each branch node, the actual integer count lies in exactly one child.

Proof Equations (2.1)–(2.8) and the order-16 specialization (4.1) put the sorted degree sequence and its forced integral triangle count in the initial list; the completeness and filtering statement is the order-16 part of Appendix A. Equations (3.1)–(3.3) follow by summing squared eigenvector coordinates over each degree class. Equations (3.4)–(3.11), after the stated elimination of H_d , follow from the diagonal of L^3 , actual neighbor sums, triangle capacities and complement counting. For a final model, take E, T, C to be the actual edge and graph/complement

triangle counts by degree class. Every row in Appendix B is then a capacity bound, double-counting identity, local third-moment identity or incidence upper bound. Finally, every branch is on an integer graph count and uses adjacent bounds $x_j \leq a$ and $x_j \geq a + 1$, so the children are disjoint and exhaustive. Thus an actual target graph remains feasible until it reaches one certified leaf. $\square$

Proof of Theorem 1

Assume that a target graph of order 16 exists. Lemma 3 maps it to an enumerated candidate and preserves a feasible point through every applicable relaxation and integer branch. Computational Proposition 1 says that the exact coverage ledger sends every such candidate to a verified Farkas leaf. The contradiction at that leaf rules out the feasible point. Hence no target graph exists. $\square$

Lemma 4 (order-20 canonical graph-to-model mapping)

Suppose that a simple graph G on 20 vertices has Laplacian spectrum $\{0, 1, \dots, 19\}$. Let G^* be G if the pair consisting of its sorted degree sequence and triangle count is the retained representative of its κ -orbit, and let $G^* = \overline{G}$ otherwise. Then G^* realizes exactly one of the 80,122 retained representatives. Its projector-diagonal weights and actual neighbor-degree sums give feasible points of its Stage-1 and Stage-2 models. If it reaches Stage 3, its actual degree-class counts E, T, C , viewed in the continuous relaxation, give a feasible point of the class model. If it is one of the final two roots, its integer $E_{2,2}$ lies in exactly one of the two children in (5.11).

Proof The target spectrum is invariant under complementation, and (5.3) is a fixed-point-free involution of the complete 160,244-candidate set. Consequently exactly one of $G, \overline{G}$ realizes the retained member of its orbit; this is the complement-coverage statement formalized in Appendix A. Applying (3.1)–(3.11) to G^* proves the Stage-1 and Stage-2 claims exactly as above. At Stage 3 take E, T, C to be the actual graph and complement counts by original degree class. The complete rows in Appendix C are satisfied by the corresponding capacity, stub, triangle-total, graph/complement third-moment, incidence and wedge counts. Although these counts are integers for G^* , the 343 ordinary class certificates discard their integrality and prove infeasibility of the larger continuous relaxation. For either final root, $E_{2,2}$ is an integer in $[0, 3]$, so it satisfies exactly one of $E_{2,2} \leq 0$ and $E_{2,2} \geq 1$. Thus G^* remains feasible until it reaches one certified leaf. $\square$

Proof of Theorem 2

Assume that a target graph of order 20 exists and form G^* as in Lemma 4. Computational Proposition 2 and the exact set-partition ledger place its retained candidate in a Stage-1, Stage-2 or Stage-3 Farkas leaf, or in one of the four certified children of the final two roots. Lemma 4 supplies a feasible point in that same leaf, contradicting its exact certificate. Hence no target graph exists. $\square$

7. Artifacts, hashes and reproduction

The theorem claims depend on two frozen proof artifacts. The canonical self-contained reproduction command from the combined release root is

```
./verification/REPRODUCE_BOTH.sh
```

It checks the top-level manifest, replays order 16, performs the hardened offline order-20 replay, and requires the reproduced order-20 canonical JSON to match the frozen expected bytes.

A successful run terminates with

COMBINED_REPRODUCTION_PASS orders=16,20

7.1 Order-16 artifact

snn16-certified-nonexistence-20260821.zip

SHA-256: f7f127da4fd6227bd66eadfc22847da270caf2d29b8bedbc16aabee548c8c847

From the extracted package root, run:

```
./REPRODUCE.sh
```

The two primary certificate files have SHA-256 identities:

certs/degree_reduction.json.gz

950d05274bf8e7234c0c72efa78927feeeacadc33ba5274f42c738ff3e8562c9

certs/class_exhaustion.json.gz

51cc5417f922f6bfceb3704c81a1691d813f76550517397269404b35225cc9bd

Expected statuses are VERIFIED_UNSAT and SEMANTIC_AUDIT_PASSED.

7.2 Order-20 artifact

S20_CERTIFIED_EXHAUSTION_RELEASE_20260821.tar.gz

SHA-256: 1eeda59a36dc835ec0efd3dc741d985145054af0d72e77f59e84f9cb63461206

To replay order 20 alone from the combined release root, run:

```
./verification/n20/REPRODUCE_INDEPENDENT.sh \  
  artifacts/S20_CERTIFIED_EXHAUSTION_RELEASE_20260821.tar.gz \  
  tmp/reproduced-n20-canonical.json
```

The publication-revision canonical JSON has SHA-256:

ebc3d8065c3f0c2e869e940f2d50c572292c45646a5ee6ac658a3f805b1ccbac

Expected statuses include:

VERIFIED_UNSAT

INDEPENDENT_ENUMERATION_V2_PASS

INDEPENDENTLY_VERIFIED_UNSAT

GRAPH_IDENTITY_CYCLE_V2_PASS

INDEPENDENT_REPRODUCTION_PASS

The strings beginning with INDEPENDENT are retained machine labels for separately implemented internal checks. They do not claim external replication.

7.3 Trust boundary

Component	Trust role
floating-point LP and generators	untrusted discovery only
analytic identities and mapping lemma	trusted mathematics
exact enumerators	trusted internal implementation
exact certificate verifiers	trusted internal implementation
coverage and branch audits	trusted internal implementation
external or different-language checker	not yet available; not claimed

The two theorem packages use different constants, candidate ledgers and final certificate structures. Passing one verifier is not treated as evidence for the other theorem.

8. Limitations and external validation

Theorems 1 and 2 concern only the two stated orders and do not settle the conjecture in general. External specialist review, independently written verifiers, clean-room reproduction on additional platforms and eventual proof-assistant formalization would provide further assurance and remain future work.

Appendix A: Exact enumeration and finite-coverage specification

This appendix specifies the discrete enumeration and coverage obligations used in the order-16 and order-20 proofs. The counts and hashes below were checked against the frozen inputs identified in Section A.1. All arithmetic in the trusted enumeration and coverage checks is integer arithmetic.

A.1 Frozen inputs and derived parameters

The two proof archives are logically separate.

snn16-certified-nonexistence-20260821.zip

SHA-256: f7f127da4fd6227bd66eadfc22847da270caf2d29b8bedbc16aabee548c8c847

S20_CERTIFIED_EXHAUSTION_RELEASE_20260821.tar.gz

SHA-256: 1eeda59a36dc835ec0efd3dc741d985145054af0d72e77f59e84f9cb63461206

For a target graph of order n , write its nondecreasing degree sequence as $s = (d_1, \dots, d_n)$. The analytic part of the proof gives the following enumeration parameters.

Parameter	$n = 16$	$n = 20$
allowed degree interval	$2, \dots, 13$	$2, \dots, 17$
sequence length	16	20
forced degree sum	120	190
forced square sum	1120	2280
constant C_n in $\sum_i d_i^3 - 6t = C_n$	11040	29260
allowed triangle interval	$54, \dots, 166$	$127, \dots, 348$
$t(G) + t(\overline{G})$	220	475

Consequently, if the allowed degree alphabet is $D_n = \{a, a + 1, \dots, b\}$, every admissible sorted sequence corresponds bijectively to a nonnegative integer multiplicity vector $(m_a, \dots, m_b)$ satisfying

$$\sum_{d=a}^b m_d = n, \quad \sum_{d=a}^b d m_d = \sigma_1, \quad \sum_{d=a}^b d^2 m_d = \sigma_2. \quad (\text{A.1})$$

No graph-isomorphism reduction or heuristic symmetry breaking is used in this enumeration.

A.2 Complete multiplicity recursion

The following pseudocode is a mathematical specification of a complete multiplicity enumeration. The frozen order-20 verifier implements its memoized exact-completion version. The separately structured order-16 audit implements the same multiplicity principle with endpoint pruning; the order-16 strict verifier instead traverses every sorted tuple directly.

```

ENUM(d, r, S, Q, multiplicities):
    # d is the smallest still available degree.
    # A completion must use r entries, with residual sum S
    # and residual square sum Q, all drawn from {d, ..., b}.

    if r < 0 or S < 0 or Q < 0:
        return

```

```

if d = b:
    if S = r*b and Q = r*b^2:
        set m_b = r
        output the sorted sequence encoded by m_a, ..., m_b
    return

if S < r*d or S > r*b:
    return
if Q < r*d^2 or Q > r*b^2:
    return
if r > 0 and r*Q < S^2:
    return
if r = 0:
    output only if S = Q = 0
    return

for c = 0, ..., r:
    set m_d = c
    ENUM(d+1, r-c, S-c*d, Q-c*d^2, multiplicities)

```

The initial call is

$$\text{ENUM}(a, n, \sigma_1, \sigma_2, ()). \quad (\text{A.2})$$

The pruning invariant is the following: at a call $\text{ENUM}(d, r, S, Q)$, a completion, if one exists, consists of exactly r values in $[d, b]$, with sum S and square sum Q . Therefore it necessarily satisfies

$$rd \leq S \leq rb, \quad rd^2 \leq Q \leq rb^2, \quad rQ \geq S^2. \quad (\text{A.3})$$

The last inequality is Cauchy–Schwarz. Hence every endpoint or Cauchy–Schwarz rejection is necessary and cannot discard a completion. At an unpruned node, the loop tries every possible value of m_d . Induction on the number of remaining degree values proves that every solution of (A.1) is output exactly once. The order-20 verifier additionally memoizes the exact Boolean recurrence

$$P(d, r, S, Q) = \bigvee_{c=0}^r P(d+1, r-c, S-cd, Q-cd^2), \quad (\text{A.4})$$

with the evident terminal condition. This can prune more nodes than (A.3), but it is an exact restatement of the same exhaustive case split.

A.3 Graphicality and forced triangle filtering

Every sequence output by (A.2) is tested for simple-graph graphicality. The strict paths use the Erdős–Gallai inequalities. If $e_1 \geq \dots \geq e_n$ is the sequence in decreasing order, the test is

$$\sum_{i=1}^k e_i \leq k(k-1) + \sum_{i=k+1}^n \min(e_i, k) \quad (1 \leq k \leq n). \quad (\text{A.5})$$

Only graphical sequences are retained. For each such sequence, the triangle count required by the third spectral moment is computed, rather than searched, as

$$t = \frac{\sum_i d_i^3 - C_n}{6}. \quad (\text{A.6})$$

The numerator must be divisible by 6, and the resulting integer must lie in the interval in Section A.1. These are necessary conditions for a target graph. Thus the moment recursion, graphicality test, divisibility test and triangle-range test cannot remove the degree data of a target graph.

A.4 Order-16 enumeration and coverage

The order-16 strict verifier enumerates all

$$d_1, \dots, d_{16} \in \{2, \dots, 13\}, \quad d_1 \leq \dots \leq d_{16},$$

by a direct combinations-with-repetition traversal. It then applies the two moment equalities, Erdős–Gallai, and (A.6). Its exact ledger is:

Order-16 layer	Source	Certified leaves	Survivors
two degree moments	all sorted tuples	–	2,249
Erdős–Gallai	2,249	–	2,233
triangle integrality and range	2,233	–	2,077
Stage 1 spectral model	2,077	1,359	718
Stage 2 local model	718	676	42
integer class trees	42 roots	108 branch-tree leaves	0

The file `certs/degree_reduction.json.gz` has one record, in exact base-list order, for each of the 2,077 candidates. Each record has exactly one of the mutually exclusive kinds `stage1_farkas`, `stage2_farkas`, or `survivor`. Writing B_{16} for the ordered base list, $L_{16,1}$ and $L_{16,2}$ for the two certified subsets, and R_{16} for the final survivors, the verifier checks

$$B_{16} = L_{16,1} \dot{\cup} L_{16,2} \dot{\cup} R_{16}, \quad (|L_{16,1}|, |L_{16,2}|, |R_{16}|) = (1359, 676, 42). \quad (\text{A.7})$$

Equivalently, the Stage-1 survivor set has size $676 + 42 = 718$, and is the disjoint union of the Stage-2 leaves and the 42 class roots. The class certificate contains exactly those 42 roots, in the same order. Its integer branch trees have 174 nodes: 66 branch nodes and 108 exact Farkas leaves, with maximum depth 10. At each branch, adjacent integer bounds $x_j \leq q$ and $x_j \geq q + 1$ are disjoint and exhaustive.

The frozen order-16 certificate identities are:

`certs/degree_reduction.json.gz`

SHA-256: 950d05274bf8e7234c0c72efa78927feeeacadc33ba5274f42c738ff3e8562c9

`certs/class_exhaustion.json.gz`

SHA-256: 51cc5417f922f6bfceb3704c81a1691d813f76550517397269404b35225cc9bd

The semantic audit follows a structurally different route: it enumerates multiplicities by the residual recursion of Section A.2 and uses largest-degree Havel–Hakimi instead of the strict verifier’s sorted-tuple traversal and Erdős–Gallai test. It reproduces 2,249, 2,233 and 2,077, compares the complete ordered base list with all certificate records, and checks that the class items equal the 42 survivors. This is a separately structured internal audit, not external replication.

A.5 Order-20 complement quotient

Before certificate elimination, the order-20 computation applies a complement quotient. For a candidate (s, t) , define

$$\kappa(s, t) = (\text{sort}(19 - d_1, \dots, 19 - d_{20}), 475 - t). \quad (\text{A.8})$$

This map preserves the complete candidate set. Indeed, complementing a simple graph realizing a graphical degree sequence gives a simple graph with degree sequence $\text{sort}(19 - d_i)$. Substitution in the first two moment equations preserves $\sum d_i = 190$ and $\sum d_i^2 = 2280$. Expanding $\sum_i (19 - d_i)^3$, and using those two moments, gives

$$\sum_i (19 - d_i)^3 = 20 \cdot 19^3 - 3 \cdot 19^2 \cdot 190 + 3 \cdot 19 \cdot 2280 - \sum_i d_i^3 = 61370 - \sum_i d_i^3.$$

Therefore (A.6) gives

$$\bar{t} = \frac{61370 - \sum_i d_i^3 - 29260}{6} = 475 - \frac{\sum_i d_i^3 - 29260}{6} = 475 - t.$$

The interval $[127, 348]$ is invariant under $t \mapsto 475 - t$. Therefore the complement mate passes the same moment, graphicality, divisibility and range predicates.

Applying (A.8) twice returns the original sorted pair, so κ is an involution. It has no fixed point: a fixed pair would require $t = 475 - t$, or $2t = 475$, which is impossible for integer t . Hence all 160,244 candidates lie in two-element orbits. Retaining the lexicographically smaller member of every orbit leaves exactly 80,122 representatives and loses no hypothetical target graph, because the graph or its complement realizes the retained member.

A.6 Order-20 enumeration, hashes and coverage

The order-20 strict route uses the exact memoized multiplicity recurrence (A.4), Erdős–Gallai, (A.6), and the complement quotient (A.8). Its ledger is:

Order-20 layer	Source	Certified leaves	Survivors
two degree moments	multiplicity recursion	–	209,932
Erdős–Gallai		–	200,108
triangle integrality and range		–	160,244
complement quotient	160,244	symmetry pairing	80,122
Stage 1 spectral model	80,122	63,235	16,887
Stage 2 local model	16,887	16,542	345
Stage 3 class model	345	343	2 roots
final integer split	2 roots	4 child leaves	0

The following SHA-256 values hash the complete **uncompressed canonical ledger bytes** named immediately above them. They are not hashes of individual certificates.

`moment_sequences.txt` (209,932)

`c60854d47d7c6492577999103aba99f00e50d076f38c3f4d3ad486e25c557856`

`graphical_sequences.txt` (200,108)

`7b7cc252551d8e0afdf76e9b47df007ac292b192c3474031201f4026936ef731`

```
base_sequences.txt (160,244)
8523a5c0c0bc1c4f2d6e76be0979f984954b393393e483f4047529b0d48245e2
```

```
complement_representatives.txt / Stage-1 source (80,122)
851b058e4c5f933e22bbd61051293342e5788fd3a524fa295266dcde4633e0d4
```

```
Stage-1 survivors / Stage-2 source (16,887)
a325883c2b1d7919f70ac82d2460733f1fe9b3e373f633b42a1b6ae512fc3ad4
```

```
Stage-2 survivors / class source (345)
dd39bed2528cea440158a0161451dc29dbbb2b36fde0f91ae6558ee7424d8798
```

```
class survivors / final roots (2)
b3b6352252492f17517e00f25307f8370daec1eb50786d598218c8f540859ecc
```

For each certificate stage i , let S_i be its ordered source ledger, R_i its ordered survivor ledger, and L_i the source indices carrying certificates. The verifier checks more than the counts: every certificate index is in range and globally strictly increasing, every survivor is unique, the survivor list is the source-induced order, and

$$S_i = L_i \dot{\cup} R_i, \quad L_i = S_i \setminus R_i. \quad (\text{A.9})$$

It also compares the bytes of each next-stage source with the preceding survivor ledger:

$$S_1 = \text{complement representatives}, \quad S_2 = R_1, \quad S_3 = R_2. \quad (\text{A.10})$$

The two Stage-3 survivors are byte-identical to the two roots in the final branch metadata. Each root is split on the actual integer graph count $E_{2,2}$ into $E_{2,2} \leq 0$ and $E_{2,2} \geq 1$. These children are disjoint and exhaustive, and all four child models have exact certificates. Thus the certificate-leaf total is

$$63235 + 16542 + 343 + 4 = 80124,$$

with zero uncovered candidates.

A.7 Structurally different internal audit paths

The agreement checks deliberately change both enumeration and graphicality algorithms.

Order	Main exact path	Structurally different internal path
16	sorted tuples by combinations with repetition; Erdős–Gallai	degree-multiplicity residual recursion; largest-degree Havel–Hakimi
20, bundled	degree-multiplicity exact-completion recursion; Erdős–Gallai	nondecreasing position-by-position feasibility dynamic program; largest-degree Havel–Hakimi
20, second verifier	does not import or execute release code	degree-alphabet meet-in-the-middle hash join; smallest-degree Kleitman–Wang layoff

The order-20 meet-in-the-middle audit splits the degree alphabet, indexes high-half multisets by `(length, sum, square sum)`, joins complementary low-half keys, and reconstructs the complete canonical bytes. It therefore does not reuse the multiplicity recursion. Its smallest-degree Kleitman–Wang layoff also differs from both Erdős–Gallai and the bundled largest-degree Havel–Hakimi path.

These checks reduce the risk of a shared algorithmic mistake, but they remain internal Python implementations using the same frozen mathematical specification and data. Section 7.3 states the corresponding trust boundary.

Appendix B: Complete computational specification for the order-16 proof

This appendix specifies the finite computation contained in the frozen order-16 archive `snn16-certified-nonexistence-20260821.zip`. The mathematical implication used throughout is

$$\text{target graph} \implies \text{feasible point of a necessary relaxation.}$$

Consequently, exact infeasibility of a relaxation rules out the graph. The converse implication is never assumed.

B.1 Frozen identities and candidate enumeration

Let G be a simple graph on 16 vertices with hypothetical Laplacian spectrum $\{0, 1, \dots, 15\}$. The spectral trace identities, the local degree bound, and the graph/complement triangle identity give

$$\sum_v d_v = 120, \quad \sum_v d_v^2 = 1120, \quad \sum_v d_v^3 - 6t = 11040, \quad (\text{B.1})$$

$$2 \leq d_v \leq 13, \quad 54 \leq t \leq 166, \quad t(G) + t(\overline{G}) = 220. \quad (\text{B.2})$$

The strict verifier enumerates every nondecreasing tuple in the lexicographic order induced by the Python standard-library combinations-with-replacement iterator over degrees $2, \dots, 13$, $s = (d_1, \dots, d_{16})$ satisfying the first two equations in (B.1). It then applies the Erdős–Gallai inequalities, and finally retains a tuple precisely when

$$t = \frac{\sum_i d_i^3 - 11040}{6} \quad (\text{B.3})$$

is an integer in $[54, 166]$. The exact ledger is

Enumeration layer	Exact count
moment tuples	2,249
Erdős–Gallai graphical tuples	2,233
graphical tuples with admissible integral t	2,077

The certificate records occur in exactly this final enumeration order. A separate internal semantic audit enumerates the moment solutions by degree multiplicities and tests graphicality by Havel–Hakimi. It reproduces all three counts and the complete ordered list of 2,077 pairs (s, t) . This is an internal cross-check using a different enumeration organization, not an external replication.

For the remainder fix one candidate (s, t) . Let

$$m_d = |\{i : d_i = d\}|, \quad D = \{d : m_d > 0\}, \quad (\text{B.4})$$

with D always traversed in increasing order.

B.2 Stage 1: spectral-weight relaxation

Choose an orthonormal Laplacian eigenbasis $u_0, \dots, u_{15}$, where u_k has eigenvalue k , and define

$$Y_{d,k} = 16 \sum_{v:d_v=d} u_k(v)^2, \quad d \in D, \quad 1 \leq k \leq 15. \quad (\text{B.5})$$

The $Y_{d,k}$ are nonnegative **real** variables. No integrality, orthostochasticity, or integral-eigenvector condition is imposed. Every target graph satisfies

$$\sum_{k=1}^{15} Y_{d,k} = 15m_d, \quad (\text{B.6})$$

$$\sum_{k=1}^{15} kY_{d,k} = 16m_d d, \quad (\text{B.7})$$

$$\sum_{k=1}^{15} k^2 Y_{d,k} = 16m_d(d^2 + d) \quad (\text{B.8})$$

for every $d \in D$, and

$$\sum_{d \in D} Y_{d,k} = 16 \quad (\text{B.9})$$

for every $1 \leq k \leq 15$. Stage 1 consists exactly of (B.6)–(B.9) and $Y_{d,k} \geq 0$. Exact rational Farkas certificates eliminate 1,359 of the 2,077 candidates, leaving 718.

B.3 Stage 2: local-third-moment relaxation

Stage 2 retains all Stage-1 variables and equations and adds one nonnegative real variable S_d for each $d \in D$. Semantically,

$$S_d = \sum_{v:d_v=d} \sum_{u \sim v} d_u. \quad (\text{B.10})$$

For one occurrence of d , delete that occurrence from s . Let ℓ_d and u_d be the sums of the d smallest and d largest entries of the remaining 15-tuple. Then

$$m_d \ell_d \leq S_d \leq m_d u_d. \quad (\text{B.11})$$

Put

$$b_d = d^3 + 2d^2, \quad K_d = \sum_{k=1}^{15} k^3 Y_{d,k}. \quad (\text{B.12})$$

There is no separately serialized H_d variable. It is the derived real quantity

$$H_d = \frac{K_d}{16} - m_d b_d = \sum_{v:d_v=d} (s_v - 2q_v). \quad (\text{B.13})$$

The exact Stage-2 inequalities are

$$m_d d \leq H_d \leq m_d d(16 - d), \quad (\text{B.14})$$

$$H_d \leq S_d \leq H_d + 2m_d \binom{d}{2}, \quad (\text{B.15})$$

and, from the local complement-triangle identity,

$$m_d \left(120 - 2 \binom{15-d}{2} \right) \leq S_d + H_d \leq 120m_d. \quad (\text{B.16})$$

The final Stage-2 equality is

$$\sum_{d \in D} S_d = 1120. \quad (\text{B.17})$$

For avoidance of sign or scaling ambiguity, the verifier serializes the six base inequality rows for each increasing d in this order:

$$-K_d \leq -16m_d(b_d + d), \quad (\text{B.18})$$

$$K_d \leq 16m_d\{b_d + d(16 - d)\}, \quad (\text{B.19})$$

$$K_d - 16S_d \leq 16m_d b_d, \quad (\text{B.20})$$

$$-K_d + 16S_d \leq 32m_d \binom{d}{2} - 16m_d b_d, \quad (\text{B.21})$$

$$-K_d - 16S_d \leq -16m_d \left\{ b_d + 120 - 2 \binom{15-d}{2} \right\}, \quad (\text{B.22})$$

$$K_d + 16S_d \leq 16m_d(b_d + 120). \quad (\text{B.23})$$

Variable-bound rows are appended later as specified in Section B.7. Exact rational Farkas certificates eliminate 676 of the 718 Stage-1 survivors. The remaining 42 candidates are the ordered records in `data/survivors42_exact.json`.

B.4 Final degree-class model

For $a, b \in D$, $a \leq b$, define the available unordered vertex-pair capacity

$$P_{a,b} = \begin{cases} \binom{m_a}{2}, & a = b, \\ m_a m_b, & a < b. \end{cases} \quad (\text{B.24})$$

For a nondecreasing degree triple $z = (a, b, c)$, let $\mu_d(z)$ be the multiplicity of d in z and define

$$P_z = \prod_{d \in D} \binom{m_d}{\mu_d(z)}. \quad (\text{B.25})$$

Only triples with $P_z > 0$ are included. The model contains:

- the nonnegative real variables $Y_{d,k}$ from Stage 1;
- $E_{a,b}$, semantically the integer number of graph edges joining degree classes a, b ;
- T_z , semantically the integer number of graph triangles with original degree multiset z ;
- C_z , semantically the integer number of complement triangles whose vertices have original graph-degree multiset z .

Thus Y is real, whereas actual graph values of E, T, C are integers. The model declares every E, T, C variable integer and imposes

$$0 \leq E_{a,b} \leq P_{a,b}, \quad 0 \leq T_z \leq P_z, \quad 0 \leq C_z \leq P_z. \quad (\text{B.26})$$

B.4.1 Stub and total-triangle equations For every $d \in D$,

$$2E_{d,d} + \sum_{b \in D \setminus \{d\}} E_{\min(d,b), \max(d,b)} = m_d d. \quad (\text{B.27})$$

The graph and complement totals are

$$\sum_z T_z = t, \quad \sum_z C_z = 220 - t. \quad (\text{B.28})$$

B.4.2 Graph classwise third moments Define

$$S_d^G = 2dE_{d,d} + \sum_{b \in D \setminus \{d\}} bE_{\min(d,b), \max(d,b)}, \quad (\text{B.29})$$

$$R_d^G = \sum_z \mu_d(z) T_z. \quad (\text{B.30})$$

For each $d \in D$, the graph classwise L^3 equation is

$$\sum_{k=1}^{15} k^3 Y_{d,k} = 16 \{m_d(d^3 + 2d^2) + S_d^G - 2R_d^G\}. \quad (\text{B.31})$$

B.4.3 Complement classwise third moments An original degree- d vertex has complement degree $15 - d$, and

$$\overline{Y}_{15-d,k} = Y_{d,16-k}. \quad (\text{B.32})$$

There are $P_{a,b} - E_{a,b}$ complement edges between original degree classes a, b . Therefore define

$$S_d^{\overline{G}} = 2(15 - d)(P_{d,d} - E_{d,d}) + \sum_{b \in D \setminus \{d\}} (15 - b) (P_{\min(d,b), \max(d,b)} - E_{\min(d,b), \max(d,b)}), \quad (\text{B.33})$$

$$R_d^{\overline{G}} = \sum_z \mu_d(z) C_z. \quad (\text{B.34})$$

The complement classwise L^3 equation is

$$\sum_{k=1}^{15} k^3 Y_{d,16-k} = 16 \left\{ m_d((15 - d)^3 + 2(15 - d)^2) + S_d^{\overline{G}} - 2R_d^{\overline{G}} \right\}. \quad (\text{B.35})$$

B.4.4 Pair-incidence inequalities For $a \leq b$, set

$$\kappa_{a,b}(z) = \begin{cases} \binom{\mu_a(z)}{2}, & a = b, \\ \mu_a(z) \mu_b(z), & a < b. \end{cases} \quad (\text{B.36})$$

This is the number of degree- (a, b) vertex pairs in a triple of type z . Every graph edge of endpoint degrees a, b lies in at most $\min(a, b) - 1$ graph triangles. Every complement edge between the same original classes has complement endpoint degrees $15 - a, 15 - b$, and hence lies in at most $14 - \max(a, b)$ complement triangles. The exact inequalities are

$$\sum_z \kappa_{a,b}(z) T_z \leq (\min(a, b) - 1) E_{a,b}, \quad (\text{B.37})$$

$$\sum_z \kappa_{a,b}(z) C_z \leq (14 - \max(a, b)) (P_{a,b} - E_{a,b}). \quad (\text{B.38})$$

B.4.5 Degree-class wedge inequalities Counting triangle incidences at vertices in one degree class gives

$$\sum_z \mu_d(z) T_z \leq m_d \binom{d}{2}, \quad (\text{B.39})$$

$$\sum_z \mu_d(z) C_z \leq m_d \binom{15 - d}{2}. \quad (\text{B.40})$$

Equations (B.6)–(B.9), (B.27)–(B.28), (B.31), and (B.35), together with (B.26) and (B.37)–(B.40), are the complete final class model. No graph isomorphism quotient or symmetry-breaking constraint is present.

B.5 Why every target graph maps into the models

For a target graph, (B.5) supplies nonnegative real Y and the spectral projector identities give (B.6)–(B.9). Actual neighbor-degree sums supply S_d ; the diagonal identity

$$(L^3)_{vv} = d_v^3 + 2d_v^2 + s_v - 2q_v \quad (\text{B.41})$$

and $s_v - 2q_v = d_v + e(N(v), R_v)$ give (B.13)–(B.15). The local complement-triangle identity gives (B.16). Actual degree-class edge and triangle counts supply integer E, T, C . Their pair and triple capacities, stub counts, triangle totals, classwise third moments, edge-triangle incidences, and wedge counts give every row in Section B.4. Complement equations follow from

$$L(\overline{G}) = 16I - J - L(G). \quad (\text{B.42})$$

Thus every target graph gives a feasible mixed point. The relaxations may also contain points that do not come from graphs; that enlargement is harmless for an infeasibility proof.

B.6 Integer branching and leaf semantics

At the root of each of the 42 final models, Y is real and E, T, C have the integer semantics and bounds in (B.26). A branch node stores a variable index j belonging to the declared integer-variable set and an integer a . Its children impose

$$x_j \leq a \quad \text{and} \quad x_j \geq a + 1. \quad (\text{B.43})$$

These children are disjoint and cover every integer value allowed at the parent. The verifier propagates current lower and upper bounds recursively and, for current bounds ℓ_j, u_j , requires $\ell_j \leq a < u_j$.

In the frozen forest all 66 branch nodes happen to branch on E -variables. This does not assume that T or C is continuous in an actual graph. At a leaf the stored Farkas certificate proves infeasibility of the **continuous** linear relaxation under the accumulated branch bounds. Hence it also excludes the smaller set in which all E, T, C are integral; no further branch is needed.

The 42 rooted trees contain

Quantity	Exact value
roots / final class models	42
all tree nodes	174
branch nodes	66
Farkas leaves	108
maximum depth	10

The full-binary-forest identity $108 = 66 + 42$ is an additional elementary check on the tree ledger.

B.7 Farkas certificates and deterministic serialization

Every model is represented as

$$Ax \leq b, \quad Cx = e. \quad (\text{B.44})$$

Finite lower bounds are appended as rows $-x_j \leq -\ell_j$; finite upper bounds are appended as $x_j \leq u_j$. A certificate consists of rational multipliers y, z such that

$$y \geq 0, \quad y^\top A + z^\top C = 0, \quad y^\top b + z^\top e < 0. \quad (\text{B.45})$$

The frozen 2,143 certificates are normalized so that the last quantity is exactly -1 . The verifier checks cancellation and negativity over fractions. Fraction, without a floating tolerance. Equation (B.45) contradicts feasibility because a feasible x would give $0 \leq -1$.

The deterministic variable order is:

1. $Y_{d,k}$, with d increasing and then $k = 1, \dots, 15$;
2. in Stage 2 only, S_d with d increasing;
3. in a class model, $E_{a,b}$ for lexicographically ordered pairs $a \leq b$;
4. T_z for lexicographically ordered nondecreasing triples with $P_z > 0$;
5. C_z in the same triple order.

The deterministic equality-row order is:

1. for each increasing d , the three rows (B.6)–(B.8);
2. for $k = 1, \dots, 15$, row (B.9);
3. in Stage 2, the single final row (B.17);
4. in a class model instead: stub rows (B.27) by increasing d , the graph and complement totals (B.28), graph L^3 rows (B.31) by increasing d , and complement L^3 rows (B.35) by increasing original d .

The deterministic base-inequality order is:

1. in Stage 2, rows (B.18)–(B.23) for each increasing d ;
2. in a class model, for each lexicographic pair $a \leq b$, graph row (B.37) followed by complement row (B.38);
3. for each increasing d , graph row (B.39) followed by complement row (B.40).

After those base inequalities, every finite lower bound is appended in variable index order, followed by every finite upper bound in variable index order. This order determines the row indices referenced by a Farkas certificate.

B.7.1 Degree-reduction JSON `certs/degree_reduction.json.gz` has top-level format `snn16-degree-reduction-v1` and fields

`base_count`, `counts`, `format`, `records`

There are exactly 2,077 ordered records. Each record contains sequence, triangles, and one of the kinds `stage1_farkas`, `stage2_farkas`, or `survivor`. The first two kinds additionally contain `cert`; a survivor contains no certificate. The exact kind counts are

```
stage1_farkas  1359
stage2_farkas   676
survivor        42
```

B.7.2 Certificate JSON A certificate is an object with arrays y and z . A nonzero rational multiplier is encoded as

`[row_index, numerator, denominator]`

with nonzero denominator. Omitted row indices mean multiplier zero. The y array indexes the complete inequality list, including bounds, and must be nonnegative. The z array indexes the equality list and is unrestricted in sign. Duplicate or out-of-range indices are rejected by the strict verifier.

B.7.3 Class-exhaustion JSON `certs/class_exhaustion.json.gz` has top-level format `snn16-class-exhaustion-v1`, count 42, and an ordered items array. Every item records its index, sequence, triangle count, statistics and root tree. A tree node has one of the forms

```
{"type": "leaf", "cert": {"y": [], "z": []}}
```

or

```
{
  "type": "branch",
  "var": 143,
  "floor": 4,
  "left": {},
  "right": {}
}
```

Both displays are schema illustrations: actual leaf multiplier arrays are populated, and the empty child objects are placeholders for recursively encoded nodes. The verifier reconstructs the model from the sequence and triangle count; no matrix stored in the certificate is trusted.

B.8 Exact coverage and frozen identities

The degree-reduction records and class forest give the disjoint coverage ledger

$$2077 = 1359 + 676 + 42, \tag{B.46}$$

$$2143 = 1359 + 676 + 108. \tag{B.47}$$

The strict verifier matches every degree record to the freshly enumerated (s, t) at the same position, matches the 42 class items to exactly the 42 survivors, checks every branch recursively, and verifies all 2,143 rational contradictions. The semantic audit independently reorganizes the finite degree-sequence enumeration, checks the survivor and complement ledgers, tests selected non-spectral graph-to-model identities on structured and deterministic random graphs, and confirms that a one-unit certificate mutation is rejected.

The frozen artifact identities are:

```
snn16-certified-nonexistence-20260821.zip
f7f127da4fd6227bd66eadfc22847da270caf2d29b8bedbc16aabee548c8c847
```

```
certs/degree_reduction.json.gz
950d05274bf8e7234c0c72efa78927feeeacadc33ba5274f42c738ff3e8562c9
```

```
certs/class_exhaustion.json.gz
51cc5417f922f6bfceb3704c81a1691d813f76550517397269404b35225cc9bd
```

```
data/survivors42_exact.json
ca899c72bf4e17c18663ee5b0d58a044347dcaad078c1a37f02bf551c1a09fa7
```

```
code/verify_certificates.py
7fdc4932f8f6a50b7bef06db8537c3eb621ae5cea9ad6910c53ea455a0458d61
```

```
code/audit_semantics.py
a6908f2f844c35cc806e356cef121ff2984662c022905276a6b910d7dcac026c
```

A clean replay from the extracted ZIP is

`./REPRODUCE.sh`

and has expected terminal statuses `VERIFIED_UNSAT` and `SEMANTIC_AUDIT_PASSED`. The strict verifier uses Python standard-library arithmetic only. The archive records successful normal and optimized Python verification; certificate regeneration is not part of the trusted replay.

The combined release additionally stores a current normal/optimized transcript:

`logs/n16-current.log`

The combined replay normalizes away elapsed time, requires the two verifier records to agree exactly, and requires the resulting summary bytes to equal:

`results/ORDER16_RESULT_FINAL.json`

B.9 Generator/verifier lineage and limits of independence

The untrusted generator uses `code/exact_models.py`, SciPy/HiGHS and SymPy. Floating-point optimization is used only to discover supports and branch choices; proposed multipliers are exactified before storage. None of the solver success or infeasibility flags is a premise of the theorem.

The trusted strict verifier is a separately implemented **internal** Python program. It uses only the standard library, re-enumerates the candidate list, reconstructs every row from the mathematical definitions, and checks the stored rational multipliers and integer branches. It does not import the generator’s `exact_models.py` or `farkas_tools.py`.

The semantic audit supplies another enumeration organization, Havel–Hakimi graphicality, selected graph-identity tests, coverage comparisons and mutation rejection. Its mutation test loads the strict verifier. Both programs are internal Python implementations; Section 7.3 states their trust boundary.

The remaining trusted boundary consists of the analytic graph-to-model mapping in Section B.5, the correctness of the internal Python verifier and runtime, and integrity of the frozen hashes. Those limitations affect the validation status of the computer-assisted proof but do not turn floating-point discovery output into a trusted premise.

Appendix C: Complete order-20 model and certificate specification

C.1 Purpose, scope and frozen artifact

This appendix gives the complete mathematical and serialization specification for the order-20 theorem in the combined manuscript. It makes the graph-to-model implication and the correspondence with the exact verifier auditable without treating generated matrices or floating-point solver output as trusted evidence.

The theorem addressed here is

there is no simple graph G on 20 vertices with $\text{spec } L(G) = \{0, 1, \dots, 19\}$.

The frozen certificate artifact is

`S20_CERTIFIED_EXHAUSTION_RELEASE_20260821.tar.gz`

SHA-256: `1eeda59a36dc835ec0efd3dc741d985145054af0d72e77f59e84f9cb63461206`

Throughout, G is a finite simple undirected graph, A is its adjacency matrix, $D = \text{diag}(d_v)$, and $L = D - A$. Every model below contains necessary conditions only. A feasible relaxation point need not come from a graph; soundness requires the other direction: every hypothetical target graph supplies a feasible point.

C.2 Global necessities, enumeration and complement quotient

The prescribed spectrum has a simple zero, so G is connected. Its first three Laplacian moments give

$$\sum_v d_v = 190, \quad \sum_v d_v^2 = 2280, \quad \sum_v d_v^3 - 6t(G) = 29260. \quad (\text{C.1})$$

Thus $|E(G)| = 95$, and the only possible triangle count for a degree sequence is

$$t(G) = \frac{\sum_v d_v^3 - 29260}{6}. \quad (\text{C.2})$$

For an orthonormal eigenbasis $Lu_k = ku_k$, put $w_{v,k} = u_k(v)^2$. Since $w_{v,0} = 1/20$,

$$\sum_{k=1}^{19} kw_{v,k} = d_v, \quad \sum_{k=1}^{19} k^2 w_{v,k} = d_v^2 + d_v, \quad \sum_{k=1}^{19} w_{v,k} = \frac{19}{20}.$$

The pointwise inequality $k^2 \leq 20k - 19$, valid for $1 \leq k \leq 19$, implies

$$20d_v^2 - 380d_v + 361 \leq 0.$$

Its integral solutions in the simple-graph range are exactly

$$2 \leq d_v \leq 17. \quad (\text{C.3})$$

For an edge uv , the number c_{uv} of common neighbors satisfies $c_{uv} \geq d_u + d_v - 20$. Hence

$$3t(G) = \sum_{uv \in E(G)} c_{uv} \geq \sum_{uv \in E(G)} (d_u + d_v - 20) = 2280 - 20(95) = 380,$$

so $t(G) \geq 127$. The identity

$$L(\overline{G}) = 20I - J - L(G)$$

shows that the target spectrum is complement-invariant. Also,

$$t(G) + t(\overline{G}) = \binom{20}{3} - \frac{1}{2} \sum_v d_v(19 - d_v) = 475. \quad (\text{C.4})$$

Applying the same lower bound to $\overline{G}$ yields

$$127 \leq t(G) \leq 348. \quad (\text{C.5})$$

Exact enumeration lists every nondecreasing 20-tuple in $[2, 17]$ satisfying the degree moments, tests graphicality, and imposes the integrality and range in (C.2) and (C.5). The counts are

$$209,932 \longrightarrow 200,108 \longrightarrow 160,244 \quad (\text{C.6})$$

for moment sequences, graphical sequences, and graphical degree-sequence/triangle-count pairs.

For $s = (d_1, \dots, d_{20})$ nondecreasing, define

$$\kappa(s, t) = (\text{sort}(19 - d_1, \dots, 19 - d_{20}), 475 - t). \quad (\text{C.7})$$

The verifier checks that κ is an involution on the complete 160,244-candidate set. It has no fixed pair because a fixed pair would require the integer equation $2t = 475$. Retaining the lexicographically smaller member of each orbit gives 80,122 representatives. If a target graph realizes a discarded member, its complement realizes the retained mate.

C.3 Stage 1: degree-class spectral weights

Let $m_d = |\{v : d_v = d\}|$, let D_s be the increasing set of degrees in a candidate s , and define nonnegative real variables

$$Y_{d,k} = 20 \sum_{v: d_v = d} u_k(v)^2, \quad d \in D_s, \quad 1 \leq k \leq 19. \quad (\text{C.8})$$

Every target graph satisfies, for each $d \in D_s$,

$$\sum_{k=1}^{19} Y_{d,k} = 19m_d, \quad (\text{C.9})$$

$$\sum_{k=1}^{19} kY_{d,k} = 20m_d d, \quad (\text{C.10})$$

$$\sum_{k=1}^{19} k^2 Y_{d,k} = 20m_d(d^2 + d), \quad (\text{C.11})$$

and, for every $1 \leq k \leq 19$,

$$\sum_{d \in D_s} Y_{d,k} = 20. \quad (\text{C.12})$$

Equations (C.9)–(C.12), together with $Y_{d,k} \geq 0$, are exactly the Stage-1 model. Orthostochasticity, eigenvector signs, and other nonlinear conditions are omitted, so this is a relaxation. Exact Farkas certificates eliminate 63,235 of the 80,122 representatives and leave 16,887.

C.4 Stage 2: local third moments and all serialized rows

For a vertex v of degree d , define

$$s_v = \sum_{u \sim v} d_u, \quad q_v = \#\{\text{triangles of } G \text{ containing } v\}, \quad h_v = s_v - 2q_v. \quad (\text{C.13})$$

Direct multiplication gives

$$(L^3)_{vv} = d^3 + 2d^2 + s_v - 2q_v = d^3 + 2d^2 + h_v. \quad (\text{C.14})$$

If $R_v = V \setminus (N(v) \cup \{v\})$, then

$$h_v = d + e(N(v), R_v), \quad d \leq h_v \leq d(20 - d). \quad (\text{C.15})$$

For a degree class put

$$S_d = \sum_{v:d_v=d} s_v, \quad Q_d = \sum_{v:d_v=d} q_v, \quad H_d = S_d - 2Q_d. \quad (\text{C.16})$$

Also put

$$A_d = \sum_{k=1}^{19} k^3 Y_{d,k}, \quad b_d = d^3 + 2d^2. \quad (\text{C.17})$$

Summing (C.14) over the class gives the elimination identity

$$A_d = 20(m_d b_d + H_d), \quad H_d = \frac{A_d}{20} - m_d b_d. \quad (\text{C.18})$$

The serialized Stage-2 model contains S_d , but it does **not** contain H_d or Q_d as variables and does not serialize (C.18) as an extra row. Those symbols only derive the six necessary inequalities below.

The graph-triangle bounds give

$$m_d d \leq H_d \leq m_d d(20 - d), \quad H_d \leq S_d \leq H_d + 2m_d \binom{d}{2}. \quad (\text{C.19})$$

For a degree- d vertex, the number of complement triangles through it is

$$\bar{q}_v = \binom{19-d}{2} - 95 + s_v - q_v. \quad (\text{C.20})$$

Consequently,

$$m_d \left(190 - 2 \binom{19-d}{2} \right) \leq S_d + H_d \leq 190m_d. \quad (\text{C.21})$$

Let

$$R_d^* = 190 - 2 \binom{19-d}{2}.$$

After substituting (C.18), the exact six serialized inequalities are

$$\begin{aligned} A_d &\geq 20m_d(b_d + d), \\ A_d &\leq 20m_d(b_d + d(20 - d)), \\ A_d - 20S_d &\leq 20m_d b_d, \\ -A_d + 20S_d &\leq 40m_d \binom{d}{2} - 20m_d b_d, \\ -A_d - 20S_d &\leq -20m_d(b_d + R_d^*), \\ A_d + 20S_d &\leq 20m_d(b_d + 190). \end{aligned} \quad (\text{C.22})$$

Delete one occurrence of d from the degree multiset and sort the remaining entries as $r_1 \leq \dots \leq r_{19}$. A degree- d neighbor set selects d distinct entries, so

$$m_d \sum_{i=1}^d r_i \leq S_d \leq m_d \sum_{i=20-d}^{19} r_i. \quad (\text{C.23})$$

Oriented-edge double counting gives

$$\sum_{d \in D_s} S_d = 2280. \quad (\text{C.24})$$

Stage 2 consists of Stage 1, real S_d variables with bounds (C.23), the six inequalities (C.22), and equality (C.24). It eliminates 16,542 of the 16,887 Stage-1 survivors and leaves 345.

C.5 Stage 3: degree-pair and degree-triple class model

Fix one of the 345 Stage-2 survivors. For $a \leq b$ in D_s , define

$$P_{a,b} = \begin{cases} \binom{m_a}{2}, & a = b, \\ m_a m_b, & a < b. \end{cases} \quad (\text{C.25})$$

For a nondecreasing triple $z = (a, b, c)$, let $\nu_d(z)$ be the multiplicity of d in z , and define

$$P_z = \prod_d \binom{m_d}{\nu_d(z)}. \quad (\text{C.26})$$

Triples with $P_z = 0$ are omitted. For an actual graph define:

- $E_{a,b}$: the number of graph edges whose endpoint-degree multiset is $\{a, b\}$;
- T_z : the number of graph triangles whose vertex-degree multiset is z ;
- C_z : the number of complement triangles whose vertices have original-graph degree multiset z .

Their elementary capacities are

$$0 \leq E_{a,b} \leq P_{a,b}, \quad 0 \leq T_z, C_z \leq P_z. \quad (\text{C.27})$$

C.5.1 Stub and triangle totals Counting degree- d edge ends gives

$$2E_{d,d} + \sum_{b \neq d} E_{\min(d,b), \max(d,b)} = m_d d. \quad (\text{C.28})$$

The graph and complement triangle totals are

$$\sum_z T_z = t, \quad \sum_z C_z = 475 - t. \quad (\text{C.29})$$

C.5.2 Graph classwise third moments Define

$$S_d^G = 2dE_{d,d} + \sum_{b \neq d} bE_{\min(d,b), \max(d,b)}, \quad (\text{C.30})$$

and

$$Q_d^G = \sum_z \nu_d(z) T_z. \quad (\text{C.31})$$

The graph classwise L^3 equation is

$$\sum_{k=1}^{19} k^3 Y_{d,k} = 20 [m_d(d^3 + 2d^2) + S_d^G - 2Q_d^G]. \quad (\text{C.32})$$

The verifier serializes (C.32) after substituting (C.30)–(C.31); those definitions are not additional rows.

C.5.3 Complement classwise third moments An original degree- d vertex has complement degree $19 - d$, and

$$\bar{Y}_{19-d,k} = Y_{d,20-k}. \quad (\text{C.33})$$

For $a \leq b$, define

$$\gamma_d(a, b) = \begin{cases} 2(19 - d), & a = b = d, \\ 19 - b, & a = d < b, \\ 19 - a, & a < d = b, \\ 0, & d \notin \{a, b\}. \end{cases} \quad (\text{C.34})$$

There are $P_{a,b} - E_{a,b}$ complement edges of original class pair (a, b) . Hence

$$S_d^{\bar{G}} = \sum_{a \leq b} \gamma_d(a, b) (P_{a,b} - E_{a,b}), \quad (\text{C.35})$$

and

$$Q_d^{\bar{G}} = \sum_z \nu_d(z) C_z. \quad (\text{C.36})$$

Put $\bar{b}_d = (19 - d)^3 + 2(19 - d)^2$. Applying the local L^3 identity to the complement and moving the $-E_{a,b}$ terms to the left gives the exact verifier row

$$\sum_{k=1}^{19} k^3 Y_{d,20-k} + 20 \sum_{a \leq b} \gamma_d(a, b) E_{a,b} + 40 \sum_z \nu_d(z) C_z = 20 m_d \bar{b}_d + 20 \sum_{a \leq b} \gamma_d(a, b) P_{a,b}. \quad (\text{C.37})$$

The $E_{a,b}$ coefficient is positive because the complement uses $P_{a,b} - E_{a,b}$.

C.5.4 Pair-incidence and wedge inequalities For $a \leq b$, define

$$\pi_{a,b}(z) = \begin{cases} \binom{\nu_a(z)}{2}, & a = b, \\ \nu_a(z) \nu_b(z), & a < b. \end{cases} \quad (\text{C.38})$$

The graph edge-triangle incidence bound is

$$\sum_z \pi_{a,b}(z) T_z \leq (\min(a, b) - 1) E_{a,b}. \quad (\text{C.39})$$

An original nonedge of class (a, b) is a complement edge with endpoint degrees $19 - a, 19 - b$, so

$$\sum_z \pi_{a,b}(z) C_z \leq (18 - \max(a, b))(P_{a,b} - E_{a,b}). \quad (\text{C.40})$$

The graph and complement wedge bounds are

$$\sum_z \nu_d(z) T_z \leq m_d \binom{d}{2}, \quad \sum_z \nu_d(z) C_z \leq m_d \binom{19-d}{2}. \quad (\text{C.41})$$

Stage 3 comprises spectral rows (C.9)–(C.12), bounds (C.27), equations (C.28)–(C.29), the substituted classwise rows (C.32) and (C.37), and inequalities (C.39)–(C.41).

C.5.5 Continuous-relaxation and integrality semantics For an actual graph, Y is generally real, whereas E, T, C are integer graph counts. In each of the 343 ordinary Stage-3 leaves, the verifier allows E, T, C to be real. Each ordinary certificate therefore proves infeasibility of a continuous relaxation containing every actual mixed graph point. Integrality metadata is not imposed in those root LPs. It is consulted only to validate the final branch variable. Root certificates eliminate 343 of the 345 models and leave two roots.

C.6 Exact Farkas convention

Before standardization, write a model as

$$A_{\text{eq}} x = b_{\text{eq}}, \quad A_{\text{ub}} x \leq b_{\text{ub}}, \quad \ell_j \leq x_j \leq u_j, \quad (\text{C.42})$$

where every lower bound is finite. Substitute $x = \ell + x'$, $x' \geq 0$. For each inequality add a nonnegative slack s ; for each finite upper bound add a nonnegative slack r :

$$\begin{aligned} A_{\text{eq}} x' &= b_{\text{eq}} - A_{\text{eq}} \ell, \\ A_{\text{ub}} x' + s &= b_{\text{ub}} - A_{\text{ub}} \ell, \\ x'_j + r_j &= u_j - \ell_j. \end{aligned} \quad (\text{C.43})$$

This gives an integer standard form

$$Bw = h, \quad w \geq 0. \quad (\text{C.44})$$

A sparse integer row multiplier z is a contradiction certificate when

$$B^T z \geq 0, \quad h^T z < 0. \quad (\text{C.45})$$

A feasible w would instead give $h^T z = w^T B^T z \geq 0$. Equality-row multipliers may have either sign. Every model-inequality and upper-bound row has its own +1 slack column, so its multiplier must be nonnegative. The checker verifies lower-bound shifts, sparse indices, integer coefficients, slack-row signs, every component of $B^T z$, and the strict inequality $h^T z < 0$. It reconstructs the model from candidate data and trusts neither serialized matrices nor floating-point solver verdicts.

C.7 Deterministic variable, row and certificate serialization

For every candidate, D_s is increasing. Stage-1 variables are $Y_{d,k}$, first by increasing d , then by $k = 1, \dots, 19$. Stage 2 uses the same block and appends S_d in increasing degree order. The class model orders variables as:

1. $Y_{d,k}$, first by increasing d , then k ;
2. $E_{a,b}$ for lexicographically ordered pairs $a \leq b$;
3. T_z for lexicographically ordered nondecreasing triples with $P_z > 0$;
4. C_z in the same triple order.

Standard-form rows are ordered as:

1. equality rows in construction order;
2. model inequalities in construction order;
3. finite variable upper bounds in variable order.

For Stage 1 and Stage 2, equality order is the three class moments (C.9)–(C.11) for increasing d , the spectral-column equations (C.12) for increasing k , and, in Stage 2, (C.24). Stage-2 inequalities are the six rows (C.22) for each increasing d . Finite S_d upper bounds from (C.23) appear later with the variable upper-bound rows; their lower bounds are handled by the shift in (C.43).

For the class model, equality order is:

1. spectral class moments for increasing d ;
2. spectral-column equations for increasing k ;
3. stub equations for increasing d ;
4. graph and complement triangle totals;
5. graph classwise L^3 rows for increasing d ;
6. complement classwise L^3 rows for increasing d .

Class inequalities are the graph and complement pair-incidence rows for each lexicographically ordered pair, followed by graph and complement wedge rows for increasing d . Upper-bound rows then follow the E, T, C order.

Each Stage-1, Stage-2, or class certificate shard has schema

```
{
  "format": "s20-standard-form-farkas-z-v1",
  "stage": "stage1 | stage2 | class",
  "records": [
    ["source_index", [{"row_index", "integer_multiplier"}]]
  ]
}
```

The quoted field values in this display denote types, not literal values in an actual shard. Within each certificate, nonzero row indices are strictly increasing, unique, and in range. Across consecutively numbered shards, source indices are globally increasing and equal the exact source-minus-survivor index list. Zero multipliers and noninteger coefficients are rejected.

The final branch file is `certs/branch_leaves.json.gz`. For each root it binds the source sequence and triangle count to a verified variable index and name, the integer split floor, and one sparse certificate for each child. Its metadata must agree with `data/final_integer_roots.json`.

C.8 Set partition, ledger identities and exact coverage

At every certified layer,

$$S_i = L_i \dot{\cup} S_{i+1}, \quad L_i = S_i \setminus S_{i+1}. \quad (\text{C.46})$$

The exact coverage ledger is:

Layer	Source	Exact Farkas leaves	Survivors
complement representatives	80,122	–	80,122
Stage 1	80,122	63,235	16,887
Stage 2	16,887	16,542	345
Stage 3	345	343	2 roots
integer split	2 roots	4	0

The five complete uncompressed ledger-byte identities are

base candidates (160,244)

8523a5c0c0bc1c4f2d6e76be0979f984954b393393e483f4047529b0d48245e2

complement representatives / Stage-1 source (80,122)

851b058e4c5f933e22bbd61051293342e5788fd3a524fa295266dcde4633e0d4

Stage-1 survivors / Stage-2 source (16,887)

a325883c2b1d7919f70ac82d2460733f1fe9b3e373f633b42a1b6ae512fc3ad4

Stage-2 survivors / class source (345)

dd39bed2528cea440158a0161451dc29dbbb2b36fde0f91ae6558ee7424d8798

class survivors / final roots (2)

b3b6352252492f17517e00f25307f8370daec1eb50786d598218c8f540859ecc

The checker verifies ordered subset membership, uniqueness, exact set difference, source order, shard numbering, and shard hashes. It records zero uncovered cases, unknown outcomes, time-outs, or crashes.

C.9 Final $E_{2,2}$ split

The two surviving roots are

$$\begin{aligned} s^{(0)} &= (2, 2, 2, 4, 6, 6, 7, 7, 9, 9, 9, 10, 12, 12, 14, 14, 15, 16, 17, 17), \\ t^{(0)} &= 229, \end{aligned} \tag{C.47}$$

and

$$\begin{aligned} s^{(1)} &= (2, 2, 2, 5, 5, 6, 6, 8, 9, 9, 9, 10, 12, 12, 14, 14, 15, 16, 17, 17), \\ t^{(1)} &= 231. \end{aligned} \tag{C.48}$$

Each has eleven degree classes and hence $11 \cdot 19 = 209$ spectral variables. The checker verifies by name, not merely by position, that class-model variable index 209 is $E_{2,2}$. Both sequences have three degree-2 vertices, so for an actual graph

$$E_{2,2} \in \{0, 1, 2, 3\}. \tag{C.49}$$

For each root, the adjacent children are

$$E_{2,2} \leq 0 \quad \text{and} \quad E_{2,2} \geq 1. \tag{C.50}$$

The left child fixes $E_{2,2} = 0$; the right contains 1, 2, 3. The children are disjoint and exhaustive for the actual integer count. Each of the four child continuous relaxations has an exact Farkas certificate. Thus

$$63,235 + 16,542 + 343 + 4 = 80,124 \quad (\text{C.51})$$

exact leaves cover all complement representatives, with each final representative replaced by two branch leaves.

C.10 Universal graph-to-model implication

Suppose a target graph existed. Equations (C.1)–(C.5) put its degree sequence and triangle count in the complete enumeration. After complementing if necessary, it realizes a retained representative. Its eigenprojector diagonals give Y satisfying Stage 1. Its actual neighbor-degree sums give S satisfying Stage 2. Its actual edge, graph-triangle, and complement-triangle counts give E, T, C satisfying every Stage-3 equation, capacity, and incidence inequality. If it reaches a final root, its integer $E_{2,2}$ lies in exactly one child of (C.50).

The exact certificate at the corresponding leaf contradicts feasibility by (C.45), and the verified set partition ensures that no candidate avoids a leaf. Therefore no target graph exists.

C.11 Reproduction identities and provenance note

The release launcher pins the following internal helper files:

```
verification/n20/verify_enumeration_v2.py
4df5db86179076364561689d8a4e62cba67890ded650ecc354f53949492bdbd9
```

```
verification/n20/verify_certificates_v2.py
b12dfe89fa5b8562844da9f963fdc9782336a3ef29219f6d481dea71170d385e
```

```
verification/n20/audit_cycle_identity_v2.py
c59c0255ba40f7b311c614b7493b760460a622ceb09fca08ce1669230248f956
```

With those helpers, the telemetry-free canonical JSON has SHA-256

```
ebc3d8065c3f0c2e869e940f2d50c572292c45646a5ee6ac658a3f805b1ccbac
```

The combined release retains two fresh full macOS replay transcripts for this current helper set:

```
logs/n20-current-run1.log
9b7ad79994af963e0ba8acfce70d2c3f29c451fb914eae4d57a4d98010989688
```

```
logs/n20-current-run2.log
f85d291f862188b2f9b653f512ea458b971f9a4910f79cddf275a2b8f302b3ec
```

Both runs exit 0 and reproduce the canonical bytes under the offline sandbox. The expected terminal statuses are:

```
INDEPENDENT_ENUMERATION_V2_PASS
INDEPENDENTLY_VERIFIED_UNSAT
INDEPENDENT_REPRODUCTION_PASS
```

These are internal cross-check status labels; Section 7.3 states their trust boundary. Earlier wrapper identities are recorded in `ARTIFACT_PROVENANCE.md`.

Data and code availability

The archival release contains this manuscript and source, the two frozen certificate archives, their detached hashes, the hardened order-20 wrapper and internal v2 verifiers, the expected canonical order-20 JSON, exact reproduction programs, and complete replay logs. The top-level manifest records the SHA-256 digest of every deposited file. The accompanying code repository is available on [GitHub](#).

Author contributions and responsibility

Jihun Kim conceived the study, performed and verified the computations, prepared the deposited materials and wrote the manuscript. He takes responsibility for the work.

AI assistance disclosure

OpenAI ChatGPT Pro and Codex assisted with research planning, drafting, code development, test design and internal review. The author checked the mathematical claims, executed the final verification and approved the manuscript and deposited files.

Funding

This research received no external funding.

Competing interests

The author declares no competing interests.

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